

Acids, Bases and Salts (Set B) — MCQ Worksheet

Science • Chapter 2 • 25 Questions, 1 mark each — Total: 25 marks

Instructions: Choose the correct option (A / B / C / D) for each question. Each question carries 1 mark. There is no negative marking. Suggested time: 30 minutes.

Q1. Litmus is obtained from:

- (A) A flowering plant
- (B) Lichens
- (C) Algae
- (D) Fungi

Q2. An indicator that has different colours in acid, neutral, and base is:

- (A) Phenolphthalein
- (B) Methyl orange
- (C) Universal indicator
- (D) Litmus

Q3. Olfactory indicator example is:

- (A) Vanilla extract
- (B) Litmus
- (C) Phenolphthalein
- (D) Methyl orange

Q4. Acids react with metals to give:

- (A) Salt + H_2
- (B) Salt + O_2
- (C) Salt + Cl_2
- (D) Salt + H_2O

Q5. Bases react with non-metal oxides to give:

- (A) Salt + water (since non-metal oxides are acidic)
- (B) Acid + water
- (C) Metal + oxygen
- (D) No reaction

Q6. Strong acid completely:

- (A) Dissociates in water
- (B) Vapourises
- (C) Burns
- (D) Polymerises

Q7. pH 14 indicates:

- (A) Strongly acidic
- (B) Strongly basic
- (C) Neutral
- (D) Pure water

Q8. pH of lemon juice is approximately:

- (A) 2
- (B) 7
- (C) 9
- (D) 13

Q9. Acid rain forms when oxides of sulphur and nitrogen mix with:

- (A) Rainwater
- (B) Salt water
- (C) Mineral water
- (D) Distilled water

Q10. Plaster of Paris is used for:

- (A) Making casts for fractured bones
- (B) Acid-base titration
- (C) Hand sanitiser
- (D) Toothpaste

Q11. Bleaching powder is used to:

- (A) Disinfect water and bleach clothes
- (B) Make sweets
- (C) Make glass
- (D) Make iron

Q12. Washing soda is used to:

- (A) Remove hardness of water and clean
- (B) Sweeten food
- (C) Make rubber
- (D) Make sugar

Q13. When CO_2 is bubbled through limewater, the white substance formed is:

- (A) $CaCO_3$
- (B) $CaCl_2$
- (C) CaO
- (D) $Ca(HCO_3)_2$

Q14. Hydrated copper sulphate has formula:

- (A) $CuSO_4 \cdot 5H_2O$
- (B) $CuSO_4 \cdot H_2O$
- (C) $Cu(OH)_2$
- (D) CuO

Q15. An antacid generally contains:

- (A) Hydrochloric acid
(C) Sodium chloride

- (B) Magnesium hydroxide
(D) Calcium carbonate (or magnesium hydroxide)

Q16. On adding solid sodium carbonate to dilute HCl, the gas evolved is:

- (A) CO_2 , turns limewater milky
(C) Cl_2 , bleaching action

- (B) H_2 , makes pop sound
(D) O_2

Q17. Equal volumes of HCl and NaOH are mixed. The pH of mixture is approximately 7. This shows:

- (A) Equimolar neutralisation
(C) Excess of acid

- (B) Excess of base
(D) Solution becomes acidic

Q18. When dilute H_2SO_4 is added to a sodium hydroxide solution containing phenolphthalein, the pink colour:

- (A) Disappears
(C) Stays the same

- (B) Deepens
(D) Turns yellow

Q19. A baking powder is added to cakes because it produces:

- (A) CO_2 to make cake fluffy
(C) Heat

- (B) Oxygen
(D) Vapours

Q20. Common salt is used in chlor-alkali process to obtain:

- (A) NaOH, Cl_2 and H_2
(C) CaCO_3

- (B) HCl
(D) NH_3

Q21. Why is gypsum heated only up to 100°C to make plaster of Paris?

- (A) So water of crystallisation is partially removed;
complete removal makes anhydrous CaSO_4
(C) Plaster melts above 100°C

- (B) It dissolves above 100°C
(D) Heat absorbs CO_2

Q22. When a base reacts with an acid, the heat evolved is called:

- (A) Heat of neutralisation
(C) Heat of combustion

- (B) Heat of vaporisation
(D) Latent heat

Q23. Phosphorus is stored under water because:

- (A) It catches fire in air
(C) It evaporates fast

- (B) It reacts with N_2
(D) It dissolves in water

Q24. Brine solution is:

- (A) Aqueous NaCl
(C) Aqueous HCl

- (B) Aqueous NaOH
(D) Aqueous CaCl_2

Q25. Carbonates and bicarbonates react with acids to produce:

- (A) Salt + water + CO_2
(C) Salt + O_2

- (B) Salt + water + H_2
(D) No reaction